DwB Discovery Portal

A New CESSDA Portal for European Research Data Discovery

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Overview

- Supports DwB goal “equal and easy access to official microdata for the European Research Area”
  - provides “more coherent system for resource discovery of official statistics”
  - demonstrates ability to ingest metadata from multiple sources, via multiple protocols

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Scope of the portal work in WP12

- Content-wise
  - Metadata from NSIs + Archives
- Technical
  - Build prototype/beta
  - Sound, future proof methods, architecture, components
  - Standards-based
  - Extensible
  - Easy to hand over to ‘sustainability’ body

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Functional aspects of the portal

- Research data discovery (obviously)
- Provider portal, QA
- Platform for additional services

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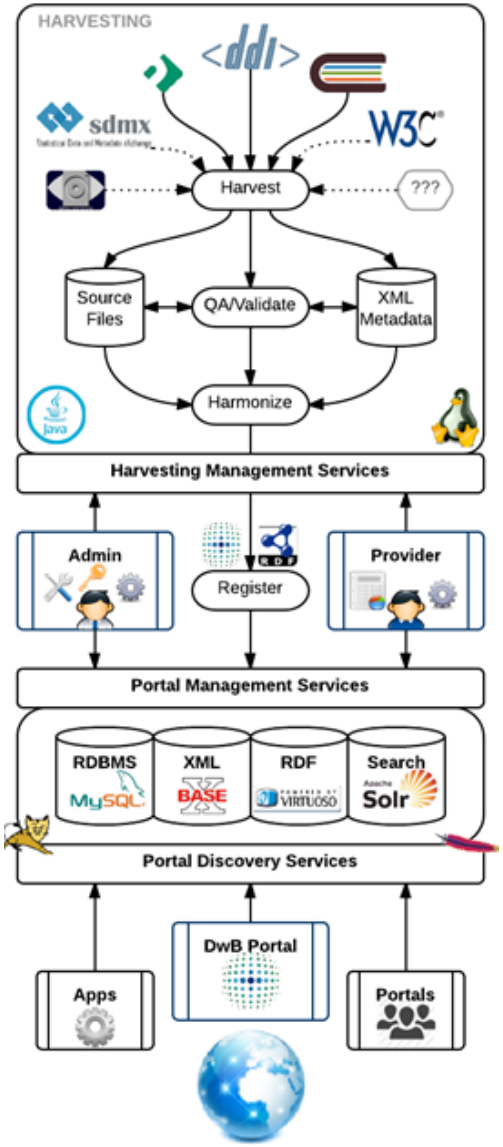
Metadata ingest

- Metadata gets...
  - harvested
  - made ready for QA
  - transformed into canonical model
  - indexed
  - exposed



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Canonical metadata model

- Harmonisation
  - DDI-C, DDI-L, MISSY, CIMES, etc.
- Builds on DISCO
  - DDI discovery RDF

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Metadata ingest dependencies

Step	Source/standard
1) Harvesting	specific
2) Produce harvesting report	specific
3) Conversion to Raw-RDF	agnostic
4) Produce conversion report	agnostic
5) Harmonization	agnostic
6) Produce harmonization report	agnostic
7) Loading	agnostic
8) Produce loading report	agnostic
9) Indexing	agnostic
10) Produce indexing report	agnostic
11+) Discovery, other downstream processes	agnostic

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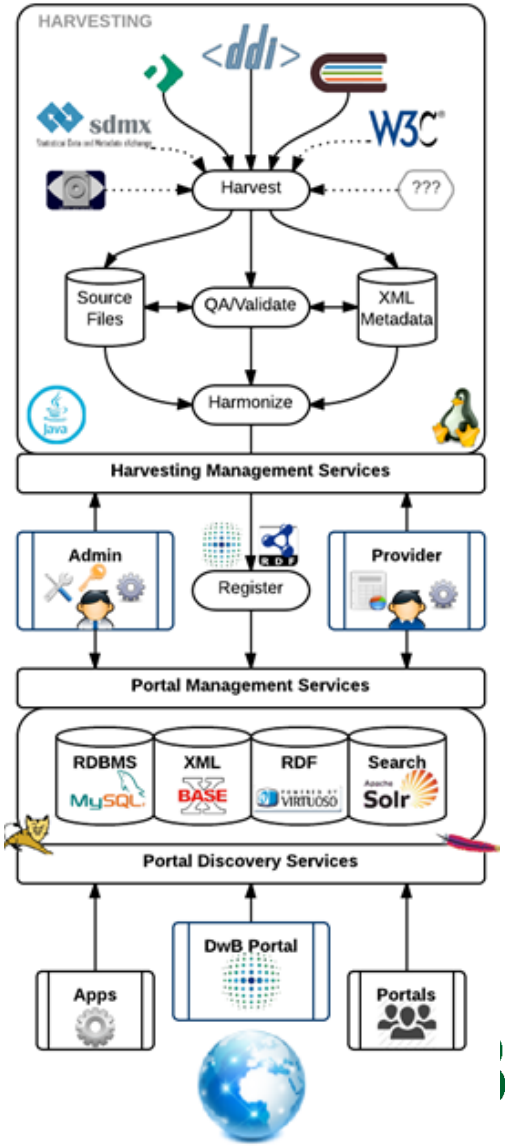
Platform for services

- DwB search portal is just a front-end application
- Machine-actionable interfaces for most functions (REST)



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Search Portal (alpha)

- Powered by Solr
- Facets
  - producer, geography, date, data type …

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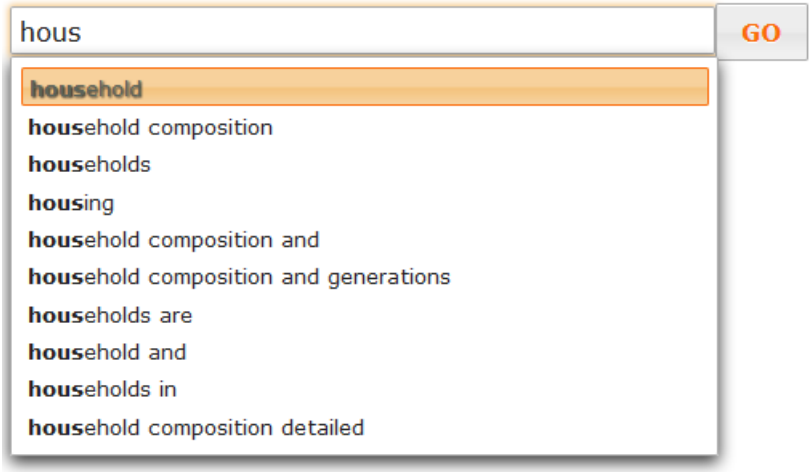
Search Portal (alpha)

- Suggestions / autocomplete



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Search Portal (alpha)

- ‘Did you mean?’ functionality

houshold GO

Did you mean?

household

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Sprint in Colchester, May 2014

- Use Jenkins CI tool to:
  - Harvest Nesstar metadata
    - any public instance
  - Load DDI XML in to BaseX
  - Convert DDI XML to raw DwB-RDF
  - Harmonize DwB-RDF
    - Simple

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Sprint in Colchester, May 2014

- Integrated Jenkins with Git to
  - Build Nesstarvester and BasexSync tools automatically
  - Update harmonization scripts automatically

- Identified mechanism to detect metadata language
  - So can check language tag is correct

- Produced Solr schema

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Jenkins Dashboard

All

S	W	Name ↓	Last Success
		Build BaseXSync via Maven	21 days - #1
		Build Nesstarvester via Maven	21 days - #28
		Config files sync	13 days - #3
		RunCaelum	15 days - #20
		RunDatabaseSync	15 days - #32
		RunHarvester	15 days - #101
		RunNesstarvesterBatch	N/A

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Jenkins Job Details

Jenkins ▸ RunHarvester ▸

Back to Dashboard

Status

Changes

Workspace

Email Template Testing

Build History (trend)

#101 20-May-2014 17:33:05

#100 20-May-2014 12:16:51

#87 15-May-2014 16:48:33

#47 06-Feb-2014 16:51:13

RSS for all RSS for failures

Project RunHarvester

Run the harvester against a data source using certain harvester type (both specif

Workspace

Recent Changes

Downstream Projects

RunDatabaseSync

Permalinks

- Last build (#101), 15 days ago
- Last stable build (#101), 15 days ago
- Last successful build (#101), 15 days ago

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Jenkins Job Details

kins ▸ RunHarvester ▸ #100

- Back to Project
- **Status**
- Changes
- Console Output
- View Build Information
- Parameters
- Git Build Data
- Previous Build
- Next Build

Build #100 (20-May-2014 12:1

No changes.

Started by user John Shepherdson

Revision: 898c6d024512e6e6a04733db22d48f44774cf4f7

- origin/master

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Metadata harmonization

- *Standard level* - sources based on the various metadata standards
- *Version level* - within a standard, the use of different versions (e.g. DDI 1.2.2, 2.5, 3.x)
- *Template/flavour level* - the use of elements of the standard for different purposes; presence/absence of optional elements
  - driven by institutional practices, templates, or software tooling

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Typical Console Output (Captured)

```
set-harvester-propfile:
     [echo] set up path to harvester properties file

update-nesstarvester-propfile:

fail-if-file-not-found:
     [echo] Checking file exists /var/dwb/config/source/uk.ukda.teaching.nv.properties
     [echo] Updating existing Nesstarvester property file path details
[propertyfile] Updating property file: /var/dwb/config/source/uk.ukda.teaching.nv.properties

run-nesstarvester:
     [echo] Running Nesstarvester process

run-jar-with-props:
     [echo] run-jar-with-props called with Jar file = ../../../lib/nesstarvester.jar
     [echo] properties file = /var/dwb/config/source/uk.ukda.teaching.nv.properties
     [echo] and class name = org.openmetadata.harvester.Harvester

fail-if-file-not-found:
     [echo] Checking file exists ../../../lib/nesstarvester.jar

fail-if-file-not-found:
     [echo] Checking file exists /var/dwb/config/source/uk.ukda.teaching.nv.properties
     [java]
     [java]
     [java]
     [java]
     [java]
     [java]
     [java] Output folder is /var/dwb/provider/uk.ukda/teaching/ddi
     [java] Starting DDI harvest on server: ukda_research
     [java]
     [java] Setting up server configuration for : http://nesstar.ukdataservice.ac.uk:80
     [java] Server configured!
     [java]
     [java] Incremental Update in Progress...
     [java]
     [java] Incremental Update Completed.
     [java]
     [java] Harvest of ukda_research completed.
     [java]
     [java] All harvest operations have been completed.
     [java] Thank you for using the OpenMetadata Nesstarvester.
     [java]

log-activity:

BUILD SUCCESSFUL
Total time: 12 seconds
Email was triggered for: Success
Sending email for trigger: Success
Sending email to: ornulf.risnes@nsd.uib.no jwshep@essex.ac.uk pascal.heus@metadatatechnology.com
Warning: you have no plugins providing access control for builds, so falling back to legacy behavior
Triggering a new build of RunDatabaseSync #31
Finished: SUCCESS
```

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What next?

- Perform provider/format specific transformations
- Apply DwB specific adjustments (identifiers, system metadata, etc.)
- Apply DwB harmonizers (map metadata in to DwB standard facets/CV etc.)
- Load harmonized DwB-RDF in to Virtuoso RDF database
- Index DwB-RDF with Solr

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What next?

- Producing various ingestion / QA reports
- Propagate deletes for survey that have been dropped
- Synchronize various metadata files to repository
  - For ‘before and after’ comparisons/provider feedback

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Any Questions?

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